

Team 10 — DineCheck (Campus Dining System)

One-Page Budget and ROI Analysis

Revised October 5, 2026. Loaded rate: \$60/hour, the instructor-provided rate for CS 4390/5388.

1. Write the assumptions

Cost Driver / Assumption	Owner
Campus dining/menu data is available and does not require a complex outside integration.	Developer
Web and mobile use one responsive application instead of separate native apps.	Developer
Crowd levels and estimated wait times remain outside this project slice.	Project manager
Testing and deployment can be completed using the team's existing development environment.	Developer
Project scope stays limited to dining locations, hours, menus, dietary filtering, and item availability.	Project manager

2. Convert and staff

From Session 9, our planning estimate is 694 team hours. This is the expected (PERT) value from our revised estimation worksheet, which gives a range of 294 to 1,150 hours.

$694 \div 130 = 5.34$ FTE-months (130 productive hours per FTE-month)

Role	Month 1	Month 2	Total
Project Manager	0.17	0.03	0.20
Developers	2.68	1.22	3.90
Testing	0.25	0.42	0.67
UX / Mobile	0.41	0.16	0.57
Total	3.51	1.83	5.34

3. Price it

$694 \text{ hours} \times \$60/\text{hour} = \$41,640$ (5.34 FTE-months at \$7,800 per FTE-month, which is \$60 $\times$ 130 hours)

Role	FTE-months	Loaded Rate	Cost
Project Manager	0.20 (26 h)	\$60/hour	\$1,560
Developers	3.90 (507 h)	\$60/hour	\$30,420

UX / Mobile	0.57 (74 h)	\$60/hour	\$4,440
Testing	0.67 (87 h)	\$60/hour	\$5,220
Labor Total	5.34 (694 h)		\$41,640

Non-Labor Costs

Item	How computed	Cost
Hosting / infrastructure	\$150/month × 2 months	\$300
Developer / Testing tools	Project allowance	\$300
Domain/deployment setup	Project allowance	\$100
Database/cloud services	Project allowance	\$300
Non-Labor Total		\$1,000

Base Build Estimate = Labor + Non-Labor = \$41,640 + \$1,000 = \$42,640

4. Price the risks

Risk	Chance	Extra hours	EMV
Menu data needs extra cleanup or access work	30%	24	\$432
Stale availability data needs more testing and rework	25%	20	\$300
Requests grow the agreed slice	20%	30	\$360
Contingency reserve			\$1,092

EMV = chance × extra hours × \$60. Chances and extra hours are team assumptions carried from our September 29 draft budget. The project manager controls contingency; the sponsor controls management reserve.

Line	How computed	Amount
Base build estimate	Labor + non-labor	\$42,640
Contingency reserve	Sum of EMV	\$1,092
Cost baseline	Base + contingency	\$43,732
Management reserve	5% of the cost baseline	\$2,187

Total budget request	Baseline + management reserve	\$45,919
-----------------------------	--------------------------------------	-----------------

Sanity check: one PERT spread from the estimate is about 143 hours (\$8,560). Contingency covers the three named risks, not a normal estimating miss.

5. Operating cost (per year, separate from the build)

Item	How computed	Cost
Hosting / infrastructure	\$150/month $\times$ 12 months	\$1,800
Maintenance	4 hours/month $\times$ 12 $\times$ \$60	\$2,880
Operating Total		\$4,680

Staff time spent marking items sold out is not included. It is an adoption risk covered in the business case.

6. ROI analysis

DineCheck does not sell anything, so the return is the value of wasted trips students no longer make. Every input below is an assumption except the wage.

Input	Value	Basis
Students who check DineCheck in a week	500	Assumption
Wasted trips avoided per user per week	0.5	Assumption (one every two weeks)
Minutes saved per avoided trip	15	Assumption, below the 25-minute walk in our interviews
Value of an hour of student time	\$7.25	U.S. federal minimum wage
Teaching weeks per year	30	Assumption (two 15-week semesters)
Benefit horizon	3 years	Assumption, as in Session 11

Return: $500 \times 0.5 \times 30 = 7,500$ avoided trips a year. $7,500 \times 15 \text{ minutes} \div 60 = 1,875$ hours $\times$ \$7.25 = \$13,594 a year, or \$40,781 over three years.

Line	How computed	Amount
Total cost, 3 years	$\$45,919 + 3 \times \$4,680$	\$59,959
Total return, 3 years	$3 \times \$13,594$	\$40,781
Net	Return – cost	–\$19,177

ROI	Net ÷ cost	−32%
------------	-------------------	-------------

Break-even: about 735 weekly users, or about 22 minutes saved per avoided trip, with the other inputs unchanged.

7. Sensitivity

Every 100 weekly users is worth about \$8,156 over three years. Build size matters as well: at the low end of the estimate (294 hours) the three-year ROI is +17%; at the high end (1,150 hours) it is −54%.

Weekly users	10 min saved	15 min saved	25 min saved
250	−77%	−66%	−43%
500	−55%	−32%	+13%
1,000	−9%	+36%	+127%

Left out on purpose: money students spend when a plan falls through and any benefit to dining services. We have no basis for averaging them yet. Sources: Session 11 Part 1, Building the Budget, slides 27 to 39; U.S. Department of Labor federal minimum wage.